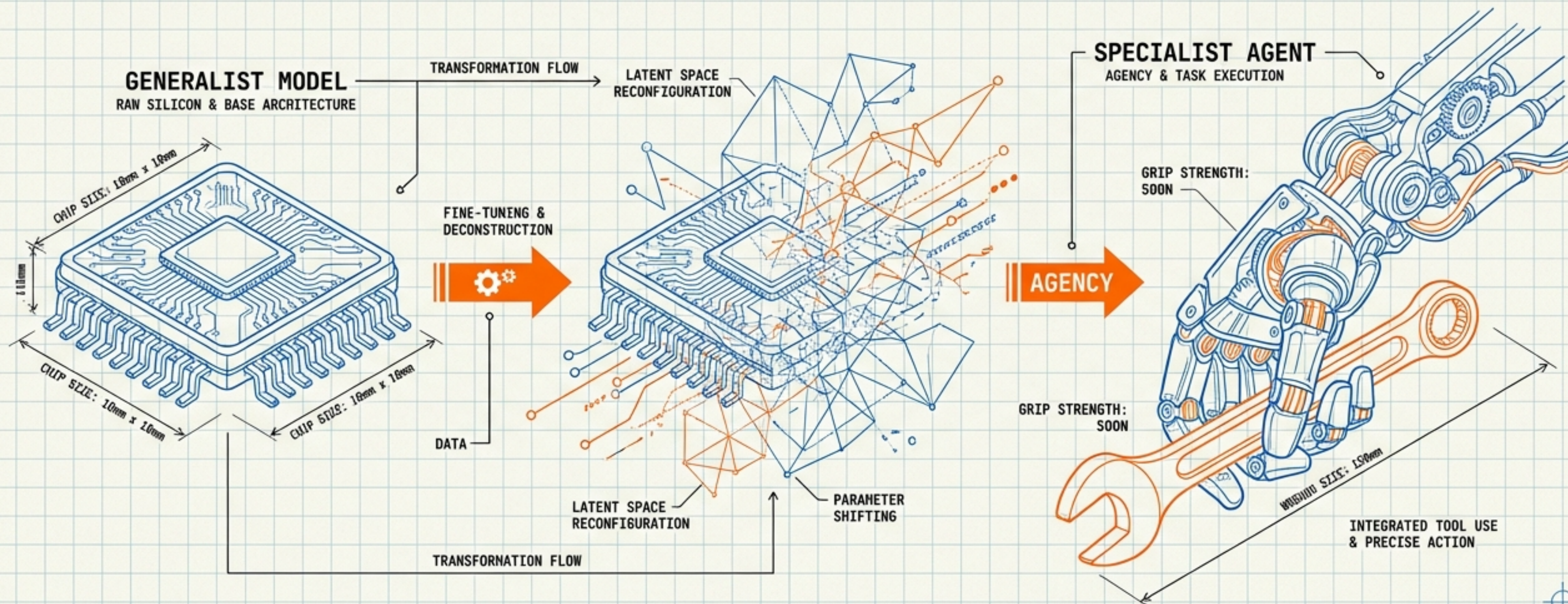
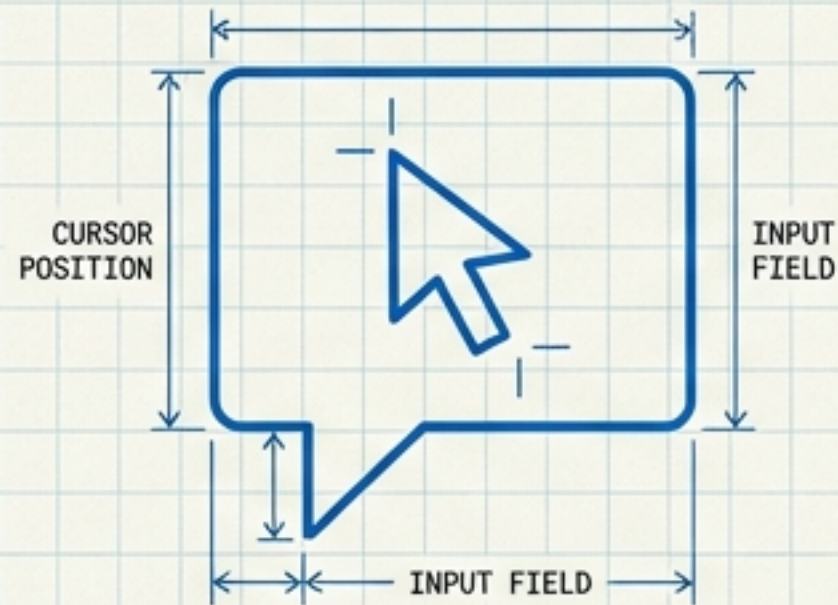


SPECIALISATION AND AGENCY: THE FINE-TUNING DEEP DIVE

From Generalist Models to Specialist Agents

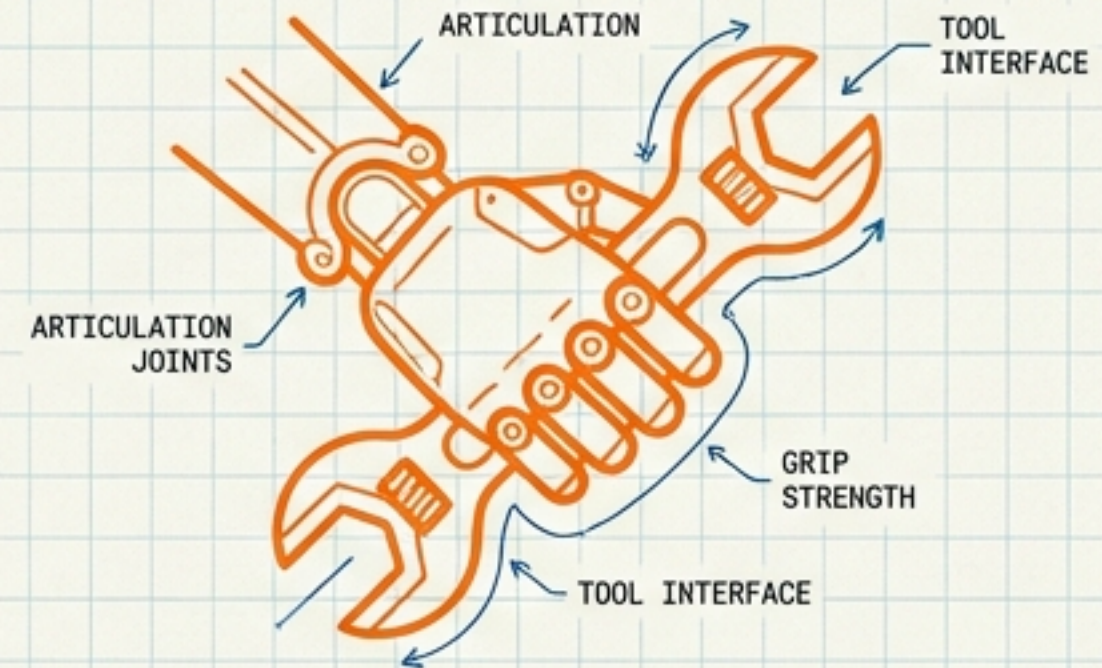


THE AWAKENING: FROM GENERATIVE TO AGENTIC



PROMPT-BASED

- Stateless ← NO MEMORY
- Single-Turn ← ISOLATED REQUEST
- Text-In / Text-Out ← NO PLANNING OF GOALS
- Context: Fire and forget.



AGENTIC AI

- Stateful ← MAINTAINS MEMORY
- Multi-Turn ← MULTI-STEP COORDINATION
- Goal-Oriented
- Context: Reasons, acts, uses tools. ← EXECUTION CAPABILITY

FUNCTIONAL SUMMARY

Agentic AI is not just about generating text; it is about architecting systems that can reason and execute.

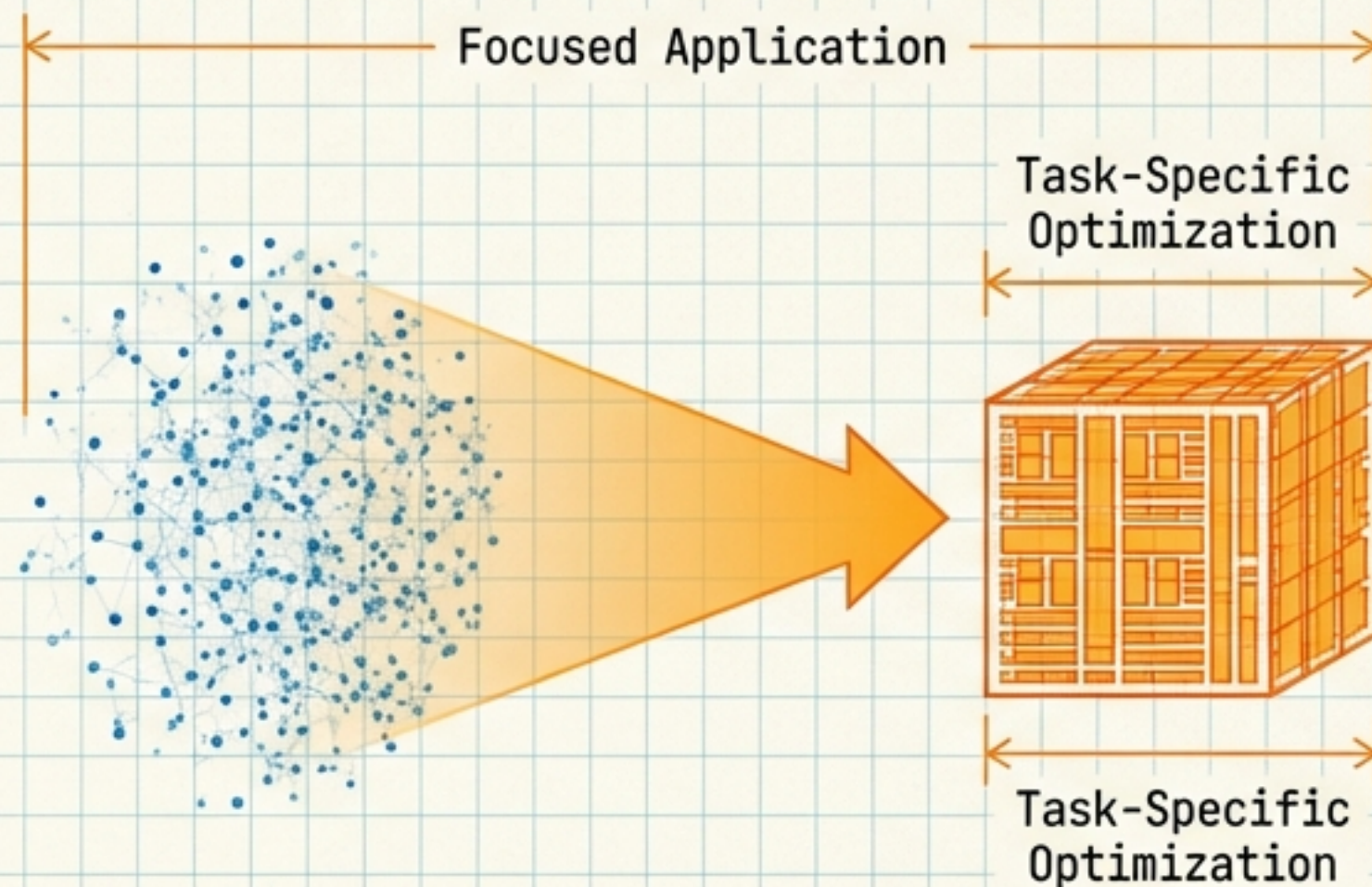
THE CONCEPT: GENERALIST VS. SPECIALIST

PRE-TRAINING (The Generalist)



Good at everything, master of nothing.
Teaches the model LANGUAGE.

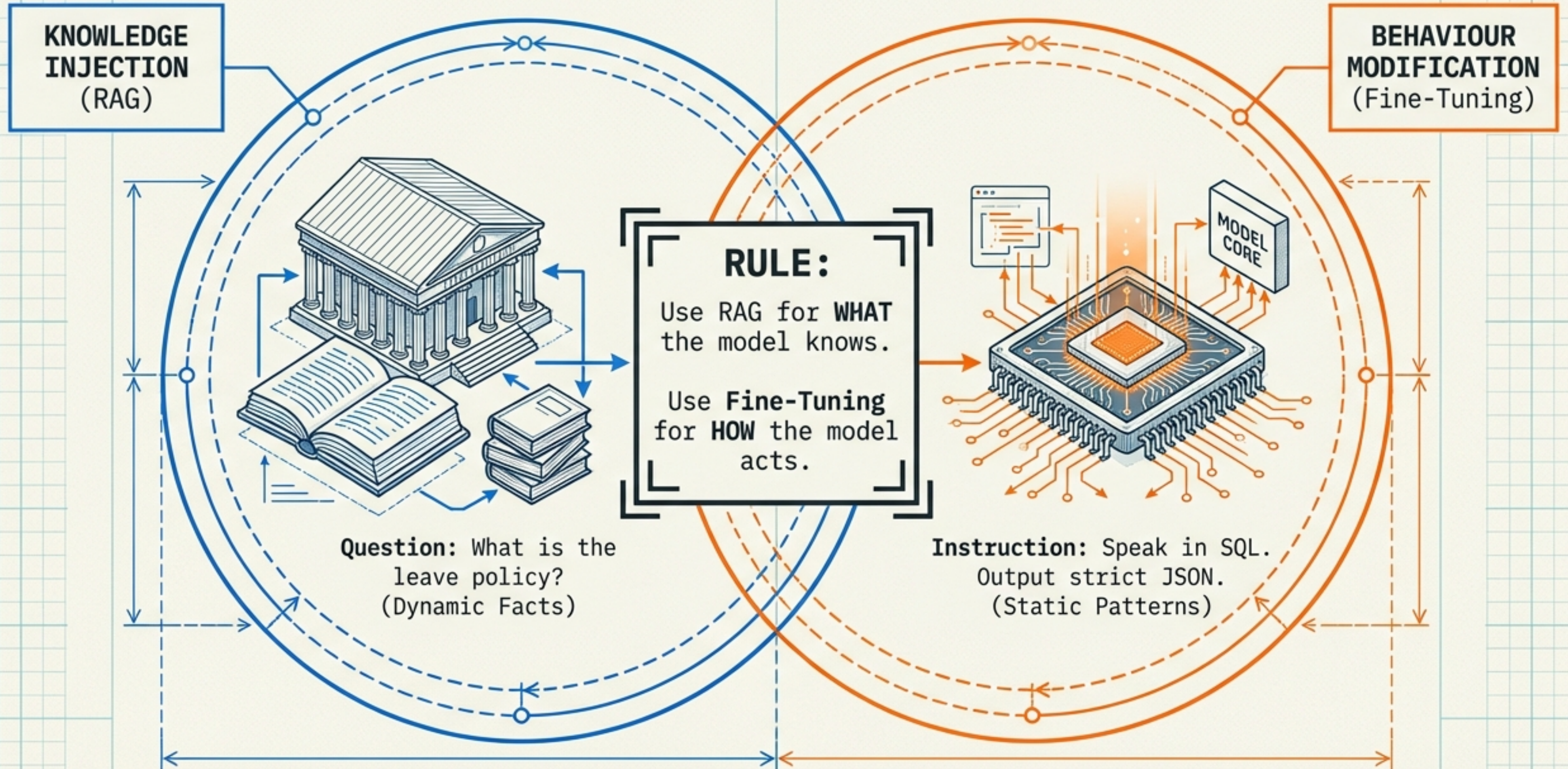
FINE-TUNING (The Specialist)



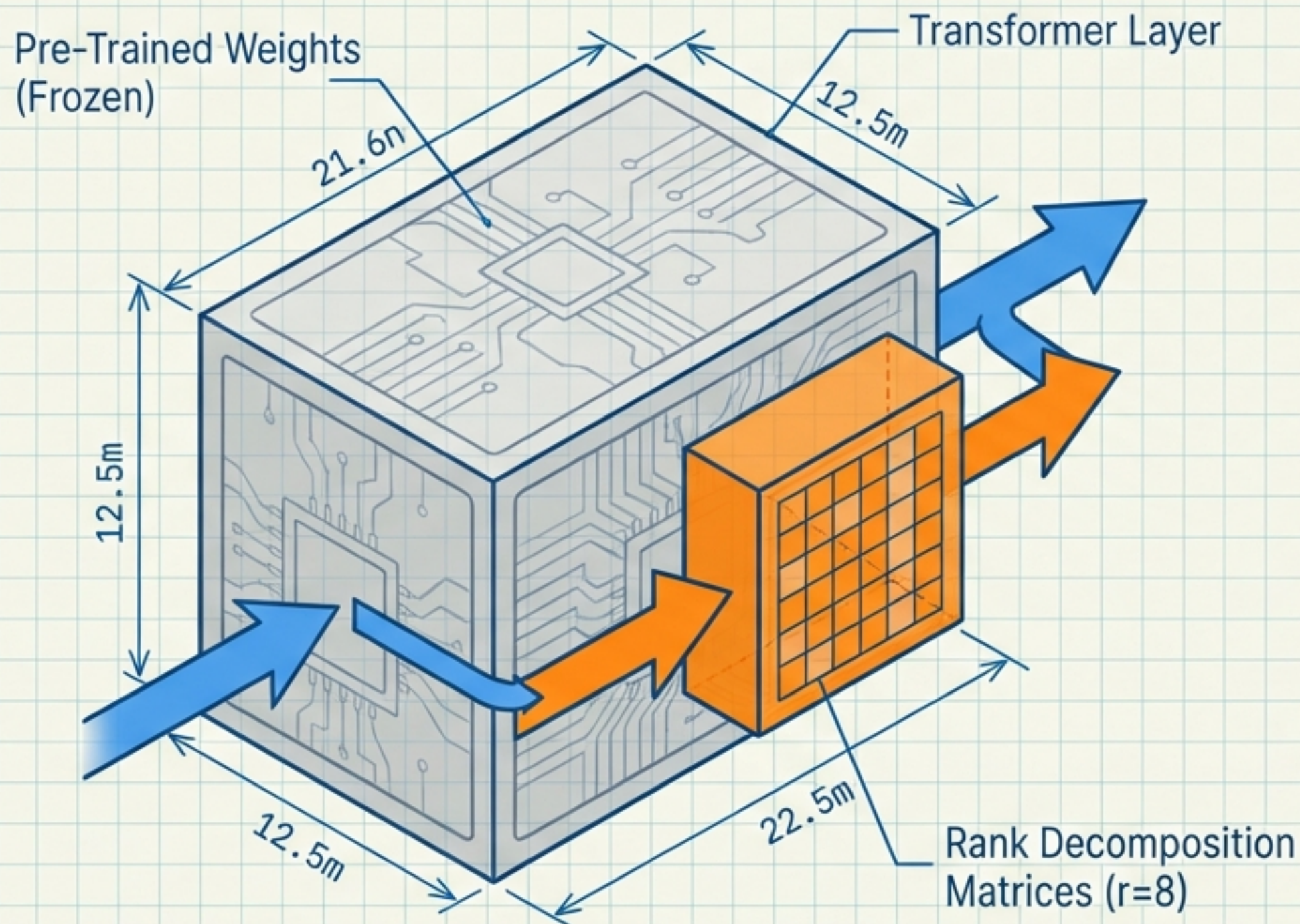
Curated dataset. Specific behaviour.
Domain expertise. Teaches the model a TASK.

Insight: We do not retrain the brain; we sharpen its focus.

THE GOLDEN RULE: ENGINEERING TRADE-OFFS



METHODOLOGY: UNDER THE HOOD (PEFT & LoRA)



PEFT (Parameter-Efficient Fine-Tuning):

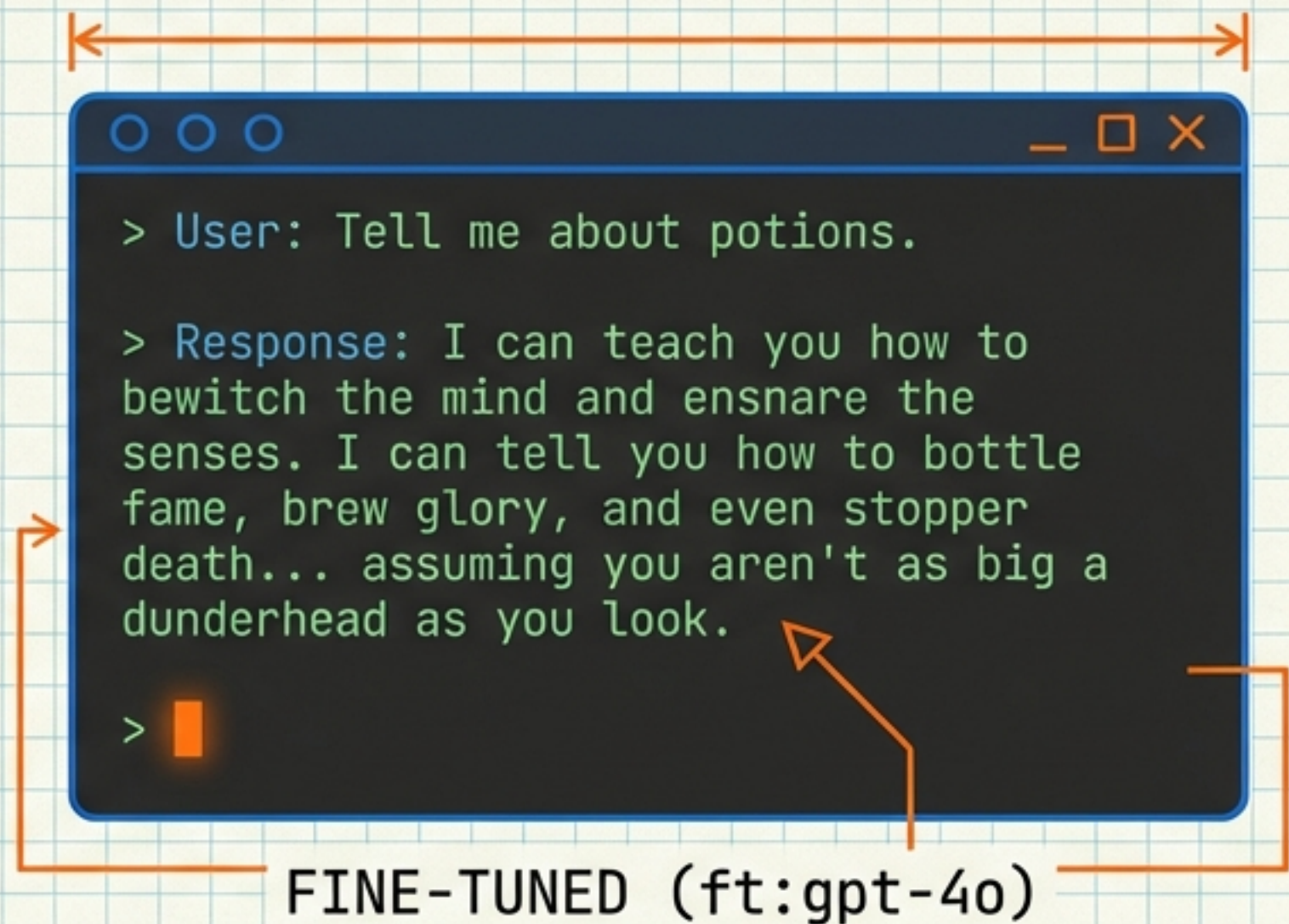
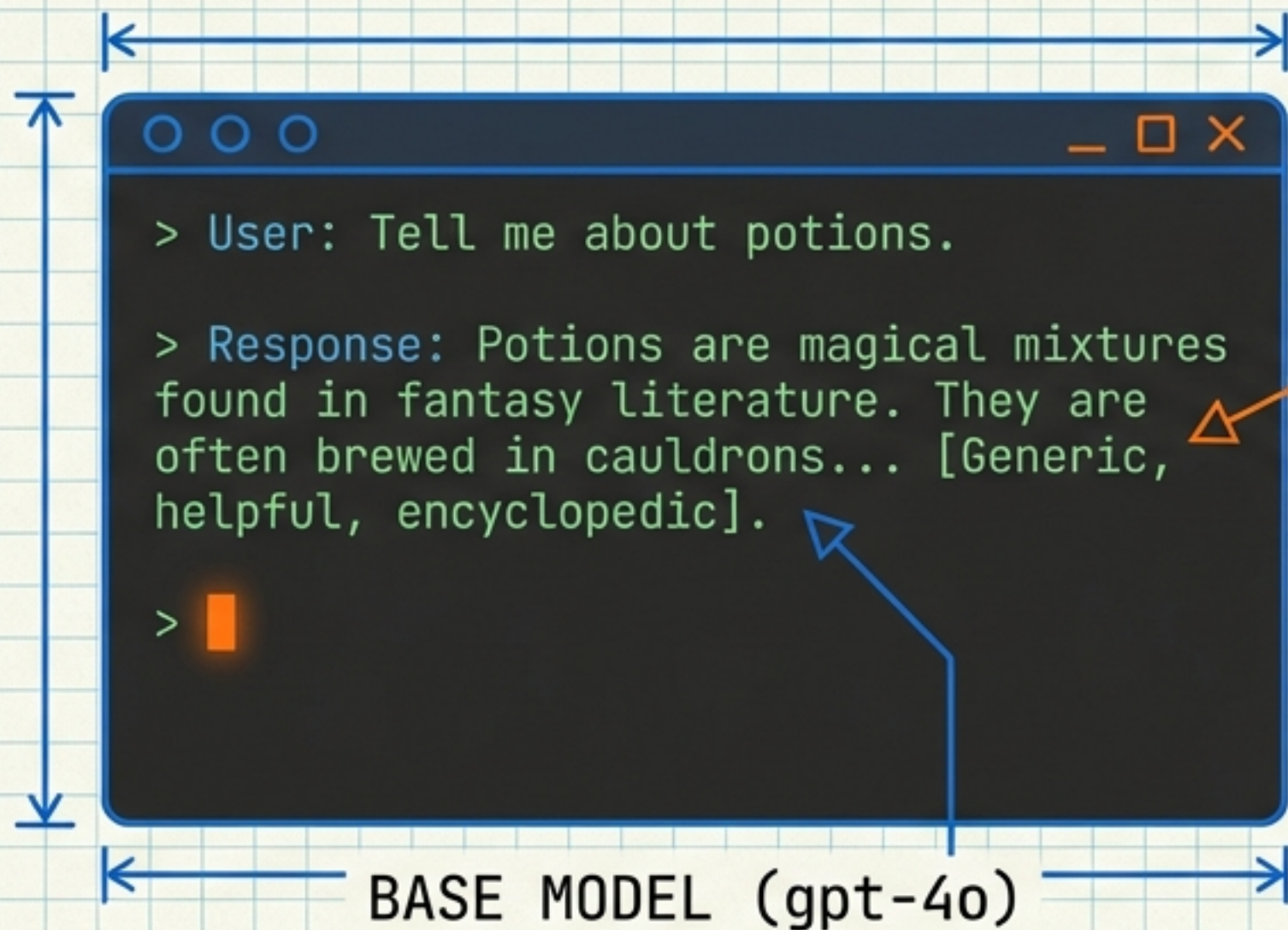
We do not retrain the entire model.
We inject small, trainable matrices.

LoRA (Low-Rank Adaptation):

Decomposes weight updates into lower-rank matrices ($r=8$), reducing trainable parameters by up to 10,000x.

CASE STUDY 1: THE PERSONA (TONE & CHARACTER)

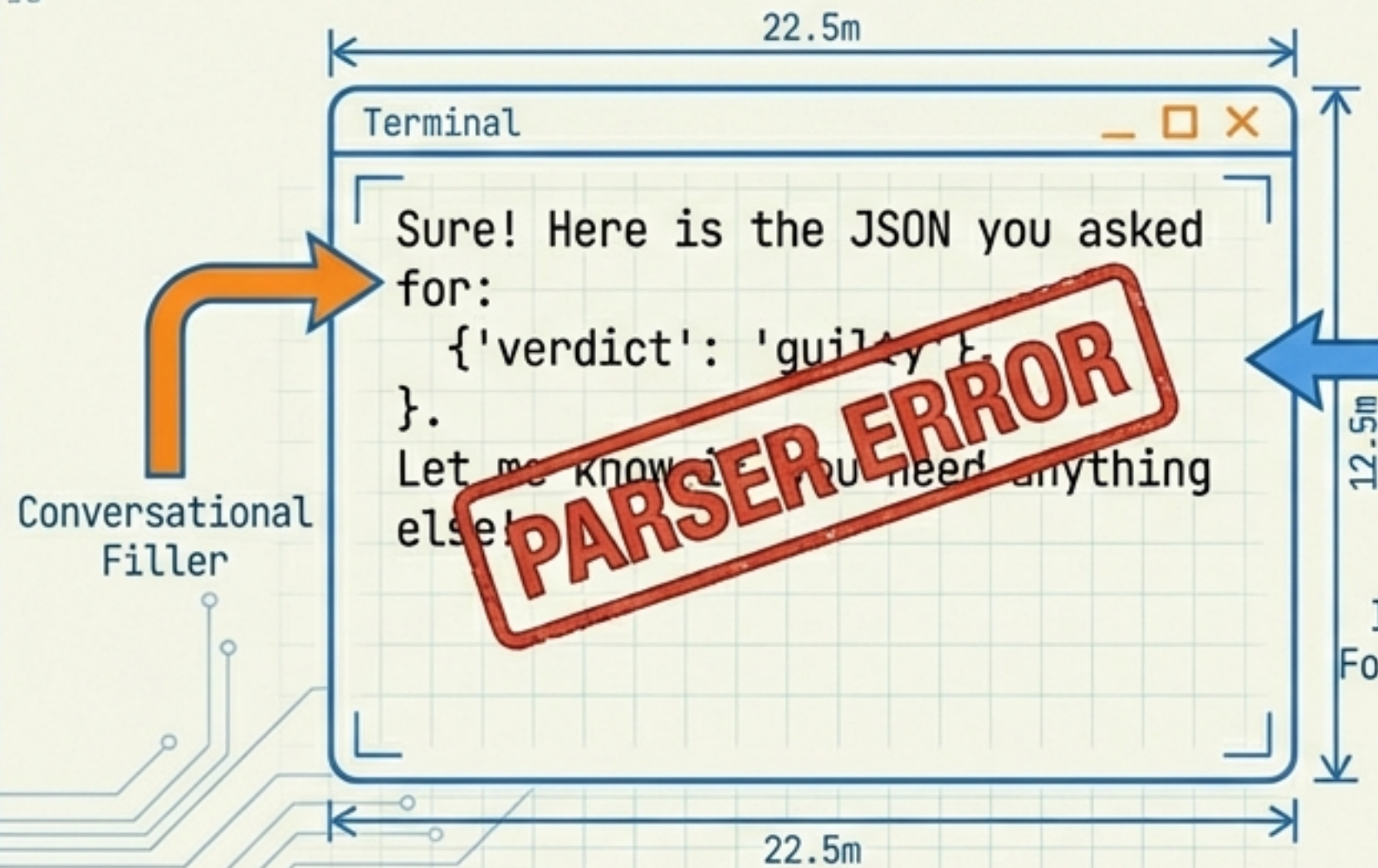
Fine-tuning for 'Professor Snape'



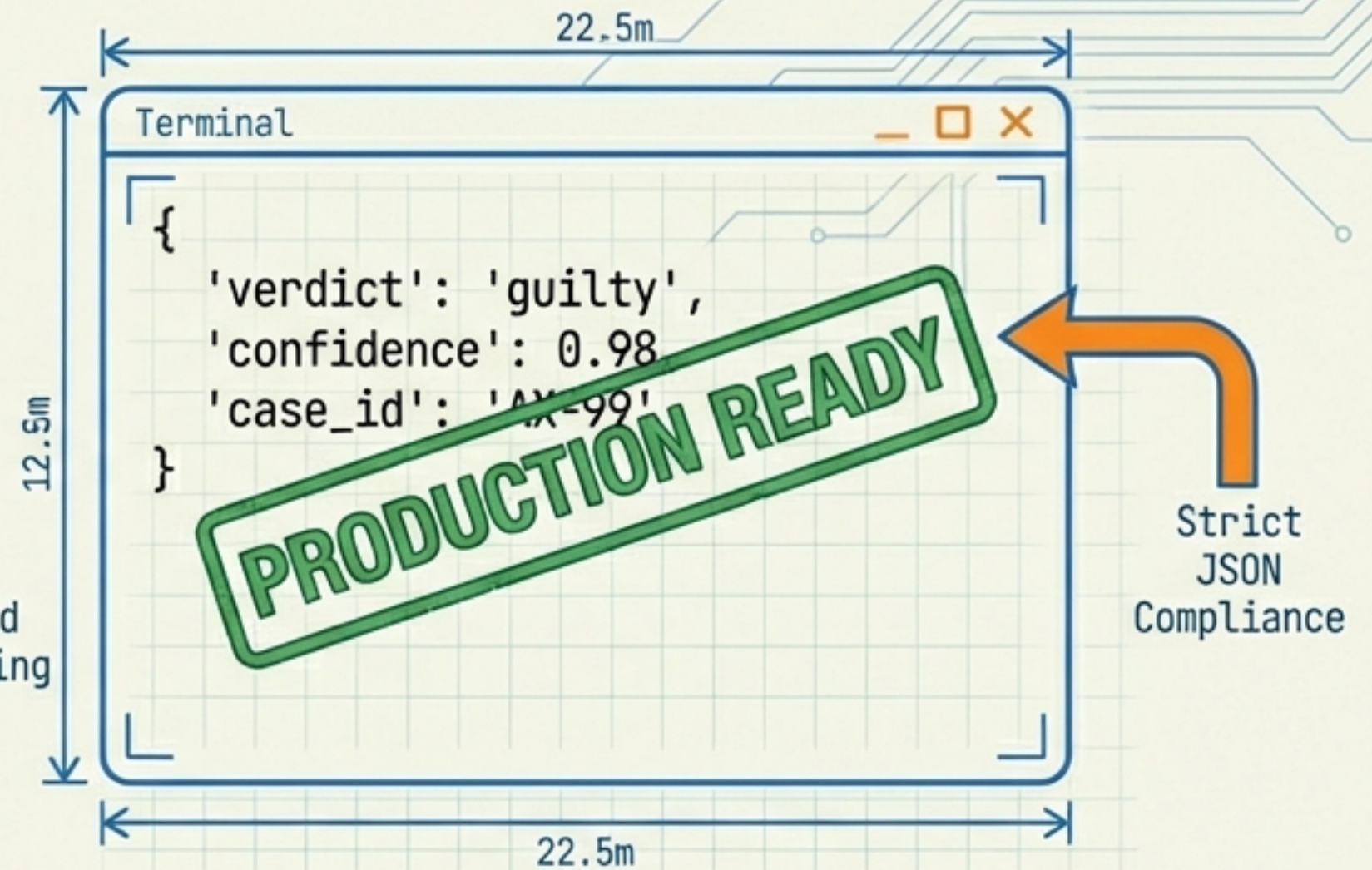
Observation: The factual knowledge is identical.
The syntax and tone are radically shifted.

CASE STUDY 2: ENFORCING STRICT SYNTAX (LEGAL TECH)

PRE-TRAINED FAILURE



FINE-TUNED SUCCESS



Fine-tuning eliminates conversational filler, ensuring machine-readable consistency.

IMPLEMENTATION: THE CODE

Standard Chat Completion

```
from openai import OpenAI
client = OpenAI()

# Generic reasoning
response = client.chat.completions.create(
    model='gpt-4o',
    messages=[{'role': 'user', 'content': 'Analyze contract.'}]
)
```

Fine-Tuned Specialist

```
# Specialized Legal Parser
response = client.chat.completions.create(
    model='ft:gpt-4o:my-org:legal-parser-v2',
    messages=[{'role': 'user', 'content': 'Analyze contract.'}]
)
```

Custom Checkpoint ID

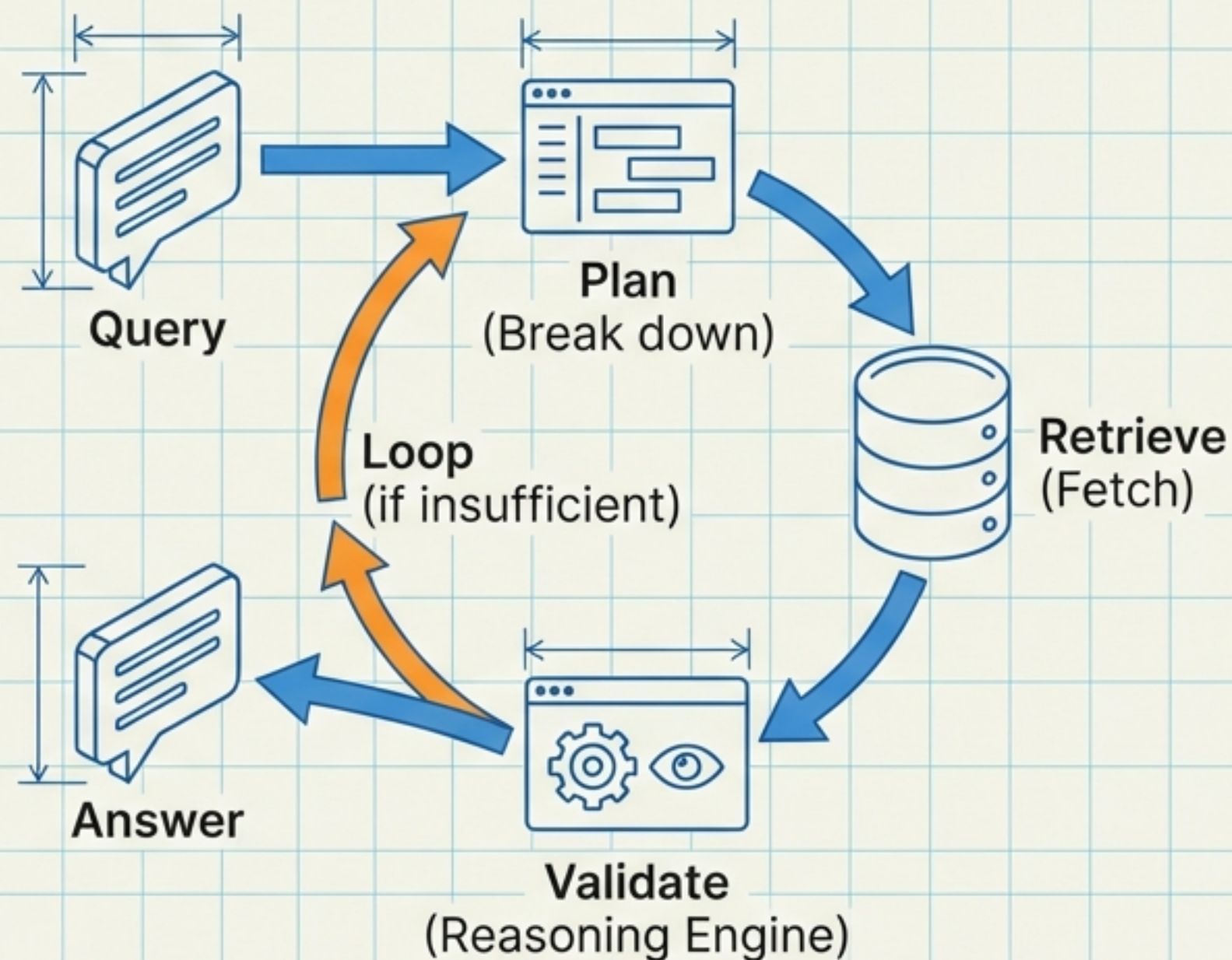
ARCHITECTURE: STATIC RAG VS. AGENTIC RAG

STANDARD RAG (Fire and Forget)



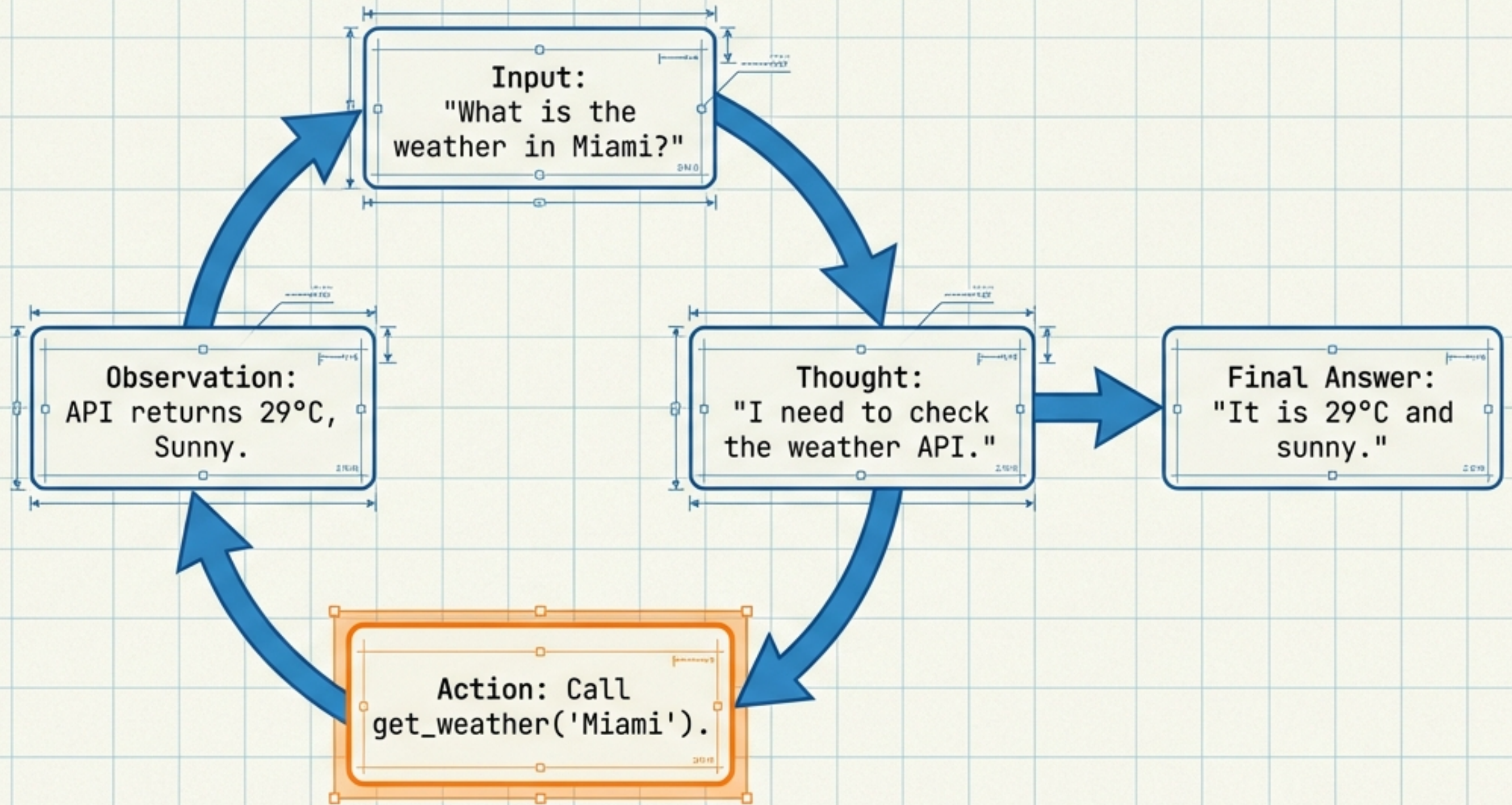
Fire and Forget.

AGENTIC RAG (Reason and Loop)



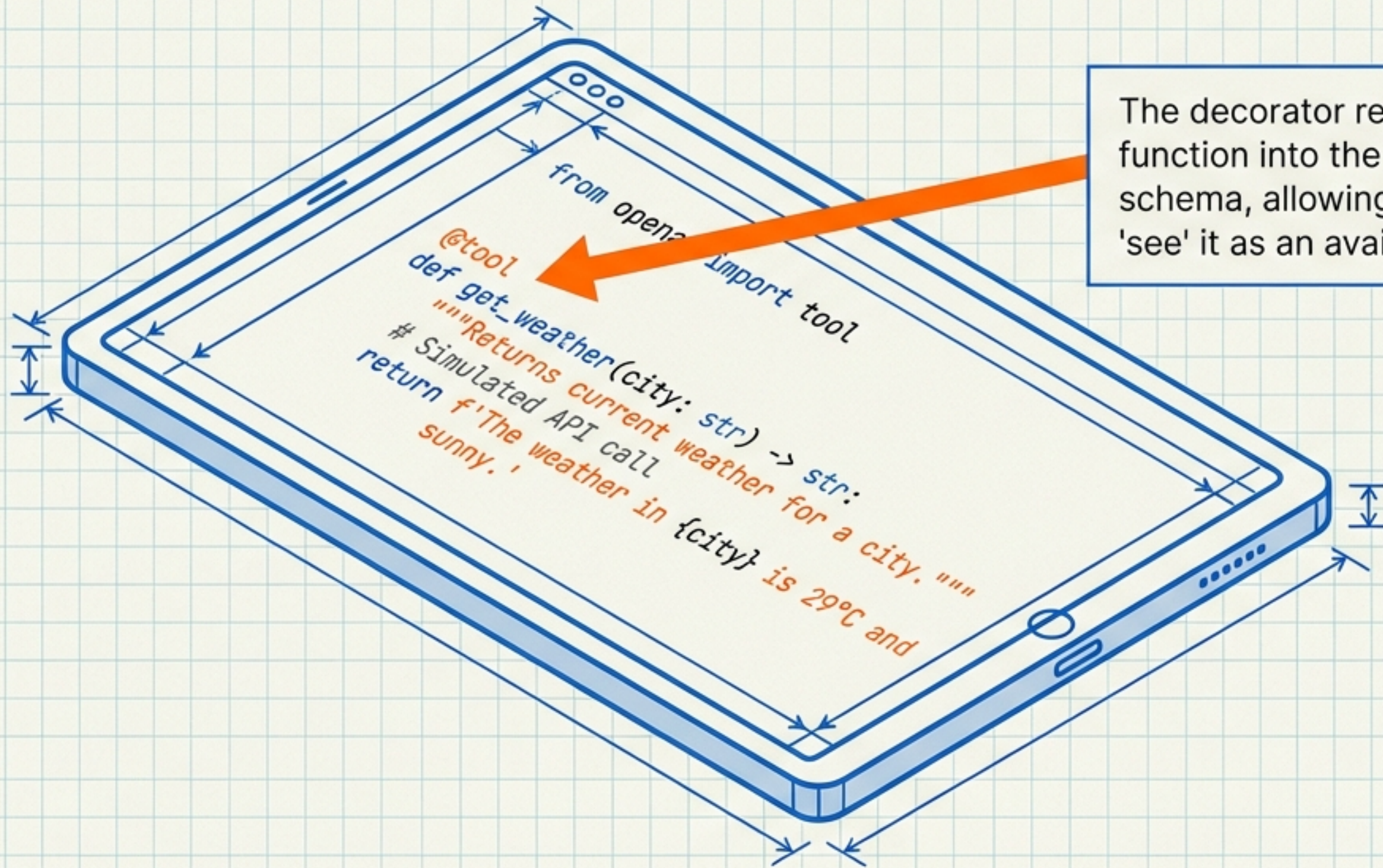
Agentic RAG evaluates data sufficiency and iterates before answering.

THE PLANNER: THE ReAct FRAMEWORK



Interleaving THOUGHT (Reasoning) with ACTION (Tool Use).

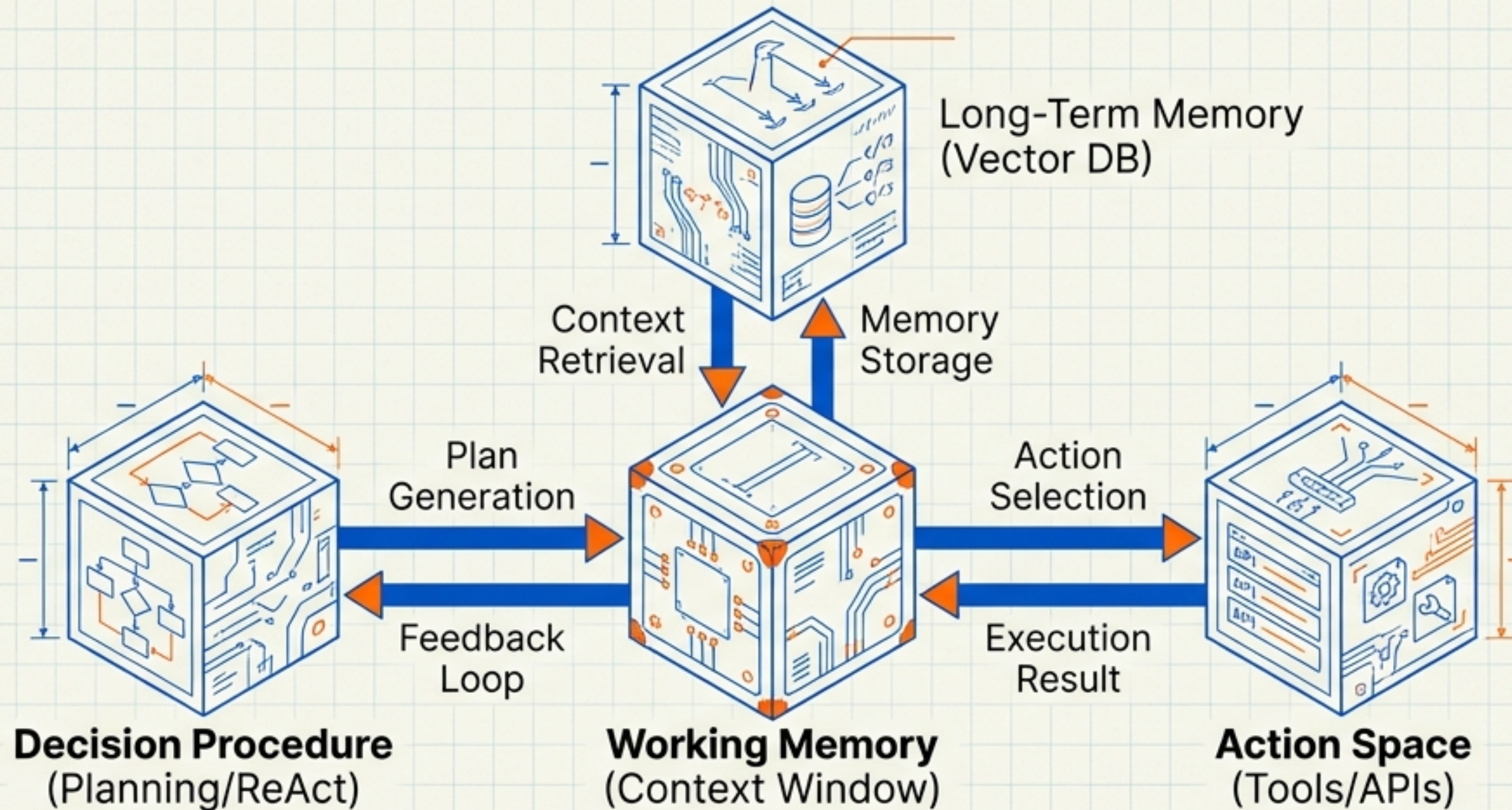
AGENT IMPLEMENTATION: DEFINING TOOLS



The decorator registers this function into the LLM's JSON schema, allowing the model to 'see' it as an available action.

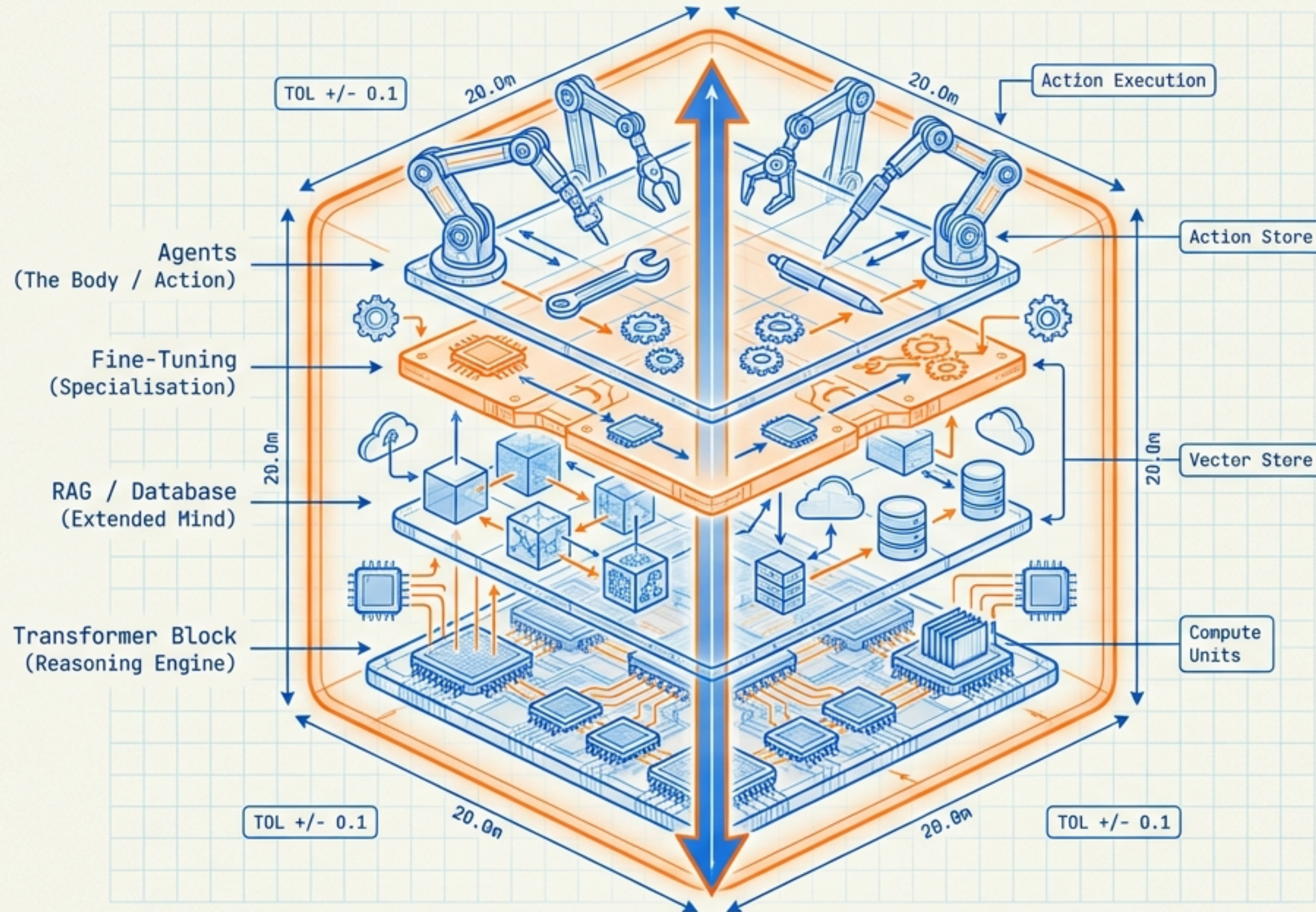
THEORETICAL FRAMEWORK: CoALA

Cognitive Architectures for Language Agents



A generalized decision-making process to choose actions based on memory and environment.

THE FULL STACK CONVERGENCE

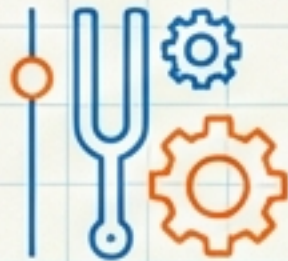


The future of AI engineering is not just in prompting models, but in architecting the cognitive loops that surround them.

SUMMARY & TAKEAWAYS

FINE-TUNING

Solves for Style, Tone, Format, and Reliability.
(The "HOW").



RAG

Solves for Factual Accuracy, Proprietary Data, and Dynamic Knowledge.
(The "WHAT").



AGENTS

Solves for Multi-step Execution, Tool Usage, and Autonomy.
(The "ACTION").



**“Don’t fine-tune for knowledge. Don’t RAG for behaviour.
Use the right tool for the layer of the stack.”**

End of Briefing.